Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination, 2024
Course Title: *Chemistry of Natural Products*
Course No.: Chem. 332; Full Marks: 70; Time: 4 hours

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[Answer any ***Five*** questions]

1. (a) Briefly discuss the isoprene and special isoprene rule with suitable examples. Mention the limitation of isoprene rule. 3
   (b) Explain the followings with necessary chemical equations:
   (i) Farnesol possess 1°–OH group. 2
   (ii) Myrcene contains three C=C double bonds, among them two of which are in conjugation. 2
   (c) How many C=C double bonds are present in limonene? Mention their positions and explain in support of it. 2
   (d) What product would you expect by the reaction of limonene with bromine in presence of carbon tetrachloride. 1
   (e) Describe all the steps involved in the synthesis of citral. 4

2. (a) Write down the chemical reactions those establishing D-glucose to contain: 2
   (i) a six carbon straight chain, and
   (ii) five –OH groups.
   (b) Discuss methylation method to determine the ring size in D-glucose. 3
   (c) Carry out the following conversions: 2+2
   (i) D-fructose to D-glucose;
   (ii) D-glucose to D-arabinose.
   (d) With necessary chemical reactions, how can you prove that the glycosidic linkage in (+)-maltose is α-1,4 between two glucose units? Draw the conformational structure of α-anomer of (+)-maltose. 3+1
   (e) Point out the major structural differences between cellulose and starch. 1

3. (a) Write down the name with structure of neutral, acidic and basic amino acids taking one from each class. 1.5
   (b) How can you synthesize phenylalanine using malonic ester synthesis method? 2.5
   (c) What happens when α-amino acids react with ninhydrin? Give reaction with mechanism. 3
   (d) Discuss Dansyl method to determine *N*-terminal amino acid of a polypeptide chain. 2.5
   (e) Give the synthesis of tripeptide, Ala-Gly-Phe using solid phase peptide synthesis. 3
   (f) What are proteins? Draw the α-helix structure of proteins. 1.5

4. (a) Write down the name and structure of nitrogeneous bases present in nucleic acids. 2.5
   (b) Give a synthesis of uric acid from urea and ethylacetoacetate. 3
   (c) What products would you expect to be formed when nucleic acids are hydrolyzed under different conditions? 3
   (d) Establish the fact that ribose or deoxyribose is in furanose form in the structure of nucleic acids. 2.5
   (e) How can you differentiate ribonucleotide-5′-phosphate from its isomers 2′- and 3′-phosphates? 1
   (f) Give an outline on the replication of DNAs. Mention the main functions of nucleic acids. 1+1

5. (a) What are alkaloids? How are the classified? 1+1=2
   (b) Discuss the extraction and isolation of alkaloids from plants. 2.5
   (c) Draw the structure of morphine and explain the followings with necessary chemical equations:
   (i) Morphine contains a phenanthrene ring. 0.5
   (ii) Morphine contains two hydroxyl groups, one of which is alcoholic and the other one is phenolic in nature. 2
   (iii) Morphine contains a >N–CH_3 group. 2
   (d) Draw two probable structures of ephedrine. Justify in support of your answer which one would be the more acceptable structure among the two. 2

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6. (a) What do you mean by bactericidal and bacteriostatic antibiotics? 2
   (b) Give the chemical name and structure of six natural penicillins. 3
   (c) Outline a method to synthesize benzylpenicillin. 3
   (d) How can you chemically establish the following facts: 1.5+1.5=3
      (i) A nitro group is present in chloramphenicol;
      (ii) Chloramphenicol contains a *para*-disubstituted benzene nucleus.
   (e) Write down the medicinal uses of chloramphenicol. 1
   (f) Point out the main causes of antibiotic resistance in our country. 2

7. (a) What are fats or oils? Draw the chemical structure. Explain clearly why fat is in solid form but oil exists as liquid form? 2
   (b) Define HDL and LDL? Which one is good for health? Explain. 2
   (c) What are steroids? Draw the basic structure and conformation of cholesterol? Show the atom numbering. 2
   (d) Apply the Blanc's rule to elucidate the different ring structures of cholesterol. Determine the positions of 2+ 3×2 = 8
      (i) –OH group
      (ii) >C=C< bond, and
      (iii) two angular $-CH_3$ groups in cholesterol.

8. (a) Write short notes on any ***four*** of the followings: 3×4=12
      (i) Extraction, isolation and purification of terpenoids;
      (ii) Hofmann's exhaustive methylation method in alkaloid chemistry;
      (iii) Mutarotation;
      (iv) Primary structure of proteins;
      (v) Gram-positive and Gram-negative bacteria.
   (b) Outline a synthesis of Diel's hydrocarbon. 2

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) *Part-III* Examination, 2023
Course Title: ***Chemistry of Natural Products***
Course No.: Chem. 332; Full Marks: 70; Time: 4 hours

[Answer any *Five* questions]

1. (a) What are terpenoids? How are they classified? 1+2
   (b) What product would you expect to be formed when citral is treated with (i) $KHSO_4$, Δ; (ii) Ag_2O and (iii) aq. K_2CO_3? 1×3
   (c) Outline a synthesis of myrcene. 2
   (d) Establish the structure of limonene by degradative and synthetic methods. 4
   (e) Show the biosynthesis of isoprene unit from mevalonic acid. 2

2. (a) Draw the primary structure of DNA and RNA. 4.0
   (b) Find out the structure of uric acid by degradative study. 3.5
   (c) Proof that the sugar molecule is attached in the five-membered ring of the purine base. 3.5
   (d) Why nucleosides are non-reducing? 1.0
   (e) How can you prove that the Medicus structure of uric acid is the actual structure? 2.0

3. (a) Explain why maltose shows mutarotation but sucrose does not. 3.0
   (b) Prove that D-glucose and D-fructose have identical configuration at C-3, C-4 and C-5 positions. 3.0
   (c) How can you synthesize D-glucose from D-arabinose? 2.0
   (d) Describe the evidence for the ring structure of aldose sugar. 3.0
   (e) How can you chemically distinguish between α-D-glucopyranose and β-D-glucofuranose? 3.0

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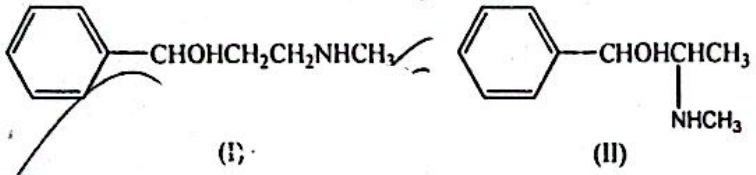
4. (a) Define antibiotics and write down the criteria for being an antibiotic. 3.0
   (b) Elucidate the structure of penicillin. 6.0
   (c) Write down different mechanisms of action, an antibiotic can follow against bacteria. 2.0
   (d) Outline the synthesis of chloramphenicol and write down some unwanted side effects of chloramphenicol. 3

5. (a) Define the terms: essential and non-essential amino acids. Give an example for each class.
   (b) How can you synthesize tryptophan from indole using hydantoin method? 2
   (c) Write equations for the reactions of phenylalanine with: 3
   (i) HNO_2, (ii) NOCl & (iii) $LiAlH_4$.
   (d) Discuss the Edman's method for determining the sequence of α-amino acids in a polypeptide chain. 4
   (e) Give a synthesis of dipeptide, gly-ala using Bergmann method. 2
   (f) What is denaturation of proteins? 1

6. (a) Define alkaloids. What are the main characters of plant alkaloids. 1+1
   (b) How can you detect the following groups in an alkaloid? 3
   (i) -OH (ii) $-OCH_2O-$ (iii) $>N-CH_3$
   (c) Establish the following facts:

(i) Ephedrine contains a mono-substituted benzene ring.
(ii) Morphine contains a phenanthrene nucleus.
(iii) Tropine contains a pyrrolidine ring.

(d) Which one would be more acceptable structure for ephedrine of the following? Give two evidences in favor of your answer. 3



(I) (II)

(e) Give a synthesis of tropic acid from acetophenone. 2

7. (a) Explain the terms (i) epimer and (ii) mutarotation with suitable examples. 2.0
(b) What is osazone? Show that the D- (+)-glucose and D-mannose give the same osazone. Explain it with proper reactions. 2.0
(c) Distinguish between reducing and non-reducing sugars with suitable examples. 2.0
(d) Write down the reactions when D-glucose reacts with the following reagents: 1×3=3.0
(i) methanol in the presence of HCl; then
(ii) dimethyl sulphate in the presence of sodium hydroxide; then
(iii) dil. HCl.
(e) Describe a method for conversion of an aldose into a higher numbered aldose. 2.0
(f) Discuss about the conformation of (+)-maltose, (+)-sucrose and starch. 3.0

8. (a) What is Diels hydrocarbon? How was the structure of Diels hydrocarbon established? 2.0
(b) What are steroids? What is the basic structural skeleton of cholesterol? Draw the chair conformation of cholesterol and mention the atom numbering. 2.0
(c) Define Blanc's rule and angular methyl group. Explain the following facts during the structure elucidation of cholesterol: 2.0 + (3×2)
(i) Determination of the sizes of A, B and D rings,
(ii) Determination of the position of the hydroxyl group and double bond, and
(iii) Determination of positions of the two angular methyl groups.
(d) What are cardiotonic glycosides and saponins? 2.0

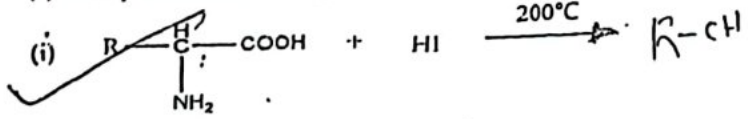
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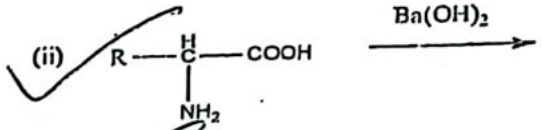
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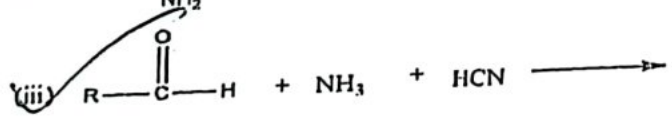
Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2022
Course Title: Chemistry of natural products
Course No. Chem-332
Time: 4 Hours, Total Marks: 70

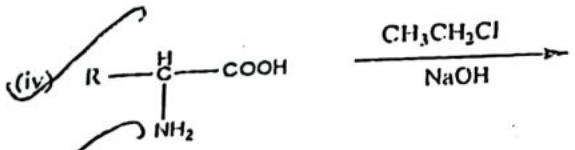
(Answer any FIVE questions)

1. (a) What is the isoelectric point of amino acid? Derive its equation with necessary conditions. 1+3=4
(b) How can you identify N-terminal and C-terminal amino acids? 2+2=4
(c) Complete the following reactions: 0.5×4=2

(i) 

(ii) 

(iii) 

(iv) 

(d) Discuss two identification tests for amino acids. 2+2=4

2. (a) Explain the extraction process of alkaloids. 3
(b) Determine the structure of tropine by degradative study. 4
(c) The two hydroxyl (-OH) groups in morphine are not phenolic. Explain with necessary reactions. 2
(d) Synthesize Ephedrine using two different methods. 3
(e) How can you identify 3° amine in cyclic form of alkaloids? 2

3. (a) Classify antibiotics based on the death and growth of cells and the number of microorganisms. 3
(b) Define Gram (+) ve and Gram (-) ve bacteria. 2
(c) Penicillin does not give a positive test for thiol (-SH) or amine (NH_2), but penicillamine does- explain with suitable reactions. 4
(d) Establish the structure of chloramphenicol antibiotic. 5

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4. (a) Define terpenoids. Briefly discuss the isoprene and special rules with examples. 1+3=4
(b) Explain the following facts: 5×2=10
(i) Myrcene contains three carbon-carbon double bonds, two of which are in conjugation.
(ii) The citral molecule has an α, β-unsaturated aldehyde group.
(iii) Ozonolysis of farnesol
(iv) The two double bonds in Limonene are not in a conjugated system; instead, one is outside the ring, and one is inside the ring.
(v) Oxidation of citral with alkaline $KMnO_4$ followed by CrO_3.

5. (a) What are purines? Why are purines so important? 2
(b) Uric acid contains imidazole ring- Explain. 4
(c) How will you synthesize adenine and guanine from uric acid? 4
(d) Discuss the Watson Crick model for DNA. 4

6. (a) Write equations showing why D-(+)- glucose and D-(+)- fructose produce the same phenylosazone. 4
(b) How can you synthesize D- glucose from D-arabinose? 3
(c) Explain how methylation and the degradation of the methylated products have helped in determining the size of the ring in D-(+)-glucose. 5
(d) Draw the conformational structures of cellulose and starch. 2

7. (a) Distinguish between lipid and phospholipid with suitable examples. 2
(b) Define sphingolipid. Draw the structures of different sphingolipids and mention the biological role of sphingolipids. 2
(c) What are steroids? What is the basic structural skeleton of cholesterol? Draw the chair conformation of cholesterol and mention the atom numbering. 2.5
(d) What is Blanc's rule? Explain it with a suitable example. 1.5
(e) Using Blanc's rule determine the sizes of different rings present in cholesterol. 4
(f) What do you mean by HDL and LDL? HDL is better than LDL for our health. Explain it. 2

8. Write short notes on any four of the following: 3.5×4=14
(a) Hoffman degradation method
(b) Replication of DNA
(c) Secondary structure of protein
(d) Position of sugar in pyrimidine base in a DNA.
(e) Structural elucidation of penicillin.

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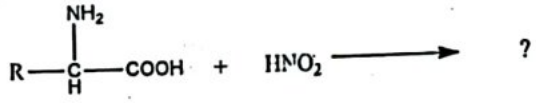
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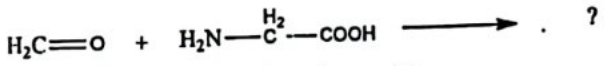
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Jahangirnagar University
B.Sc. (Hons.) Part-III Final Examination 2021
Course Title: Chemistry of natural products
Course: Chem. 332, Full Marks: 70, Time: 4 hours

(Answer any Five Questions)

1. (a) Write the structure of two imino acids, one optically active and one optically inactive amino acid. 2
   (b) Classify amino acid according to the properties of their side chain. 3
   (c) In Glycine the pKa values of amino group and carboxylic acid groups are 9.60 and 2.40 respectively, find out the isoelectric point of this amino acid. 2
   (d) Complete the following reactions: 0.5×2=1

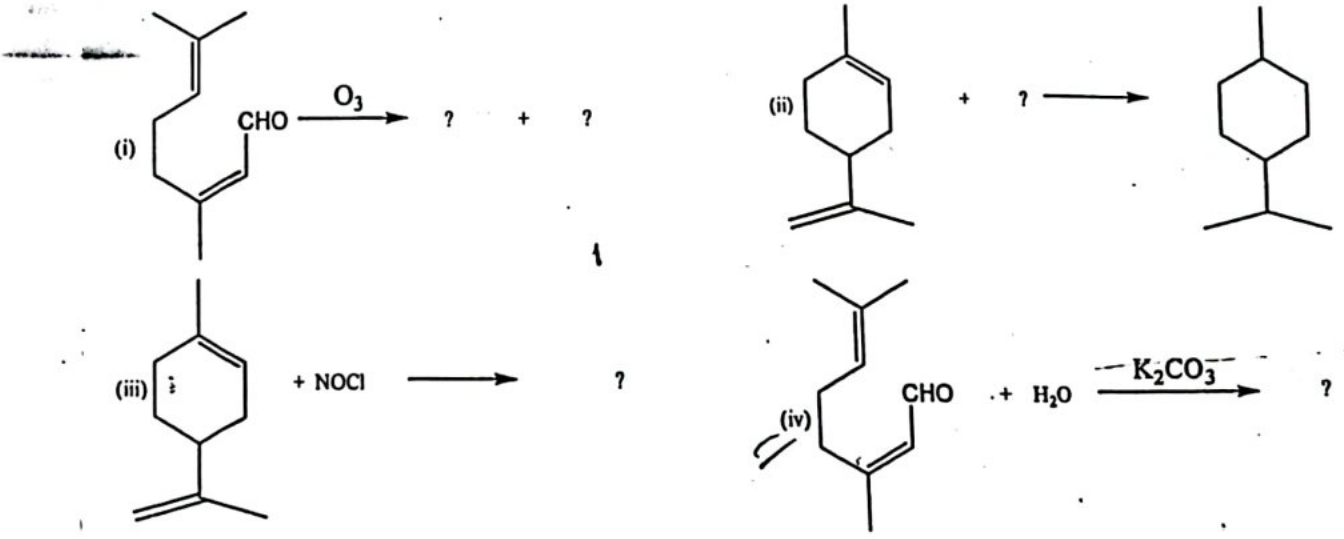




   (e) Write the two different synthetic methods of amino acid. 1.5×2=3
   (f) Write the synthesis of AlanyltyrosylGlycine peptide. 3

2. (a) Explain Herzig-Meyer method for determining N-CH_3 groups in alkaloid. 2
   (b) Why L (+)-Ephedrine is more basic than D (-) Ephedrine? 2
   (c) Ephedrine is an alkaloid successfully used for the treatment of bronchial asthma and other allergic condition. Elucidate the structure of this alkaloid. 4
   (d) Draw the structure of Morphine. 1
   (e) Prove that Atropine is an ester. 2
   (f) When Emde's degrative method is used and why? 2+1=3

3. (a) Discuss the importance of natural products. 2
   (b) Classify terpenoids on the basis of the number of isoprene unit. 3
   (c) Determine the structure of Limonene. 3
   (d) Name and draw the structure of a sesquiterpene. 2
   (e) Complete the following reactions: 4



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4. (a) Draw the structure of cholestereol with the numbering of all carbon atoms and answer the following questions:
(i) Is cholesterol a stereoid? If so, then why?
(ii) What is the functional nature of the hydroxyl group of cholesterol? Give explanation with necessary chemical equations.
(iii) "The hydroxyl group of cholesterol is present in the ring system but not in the open chain system"- justify the statement with necessary chemical reaction.
(b) Draw the structures of lecithin and cephalin? What types of lipids they are? 2.5
(c) Is there any basic difference between triglycerides and phosphoglycerides? If so, then what is the difference? 1.5
(d) Is cerebroside a phospholipid? Give explanation in favor of your answer. 1
(e) What are the LDL and HDL? Which one of them reduces the risk of heart disease and how does it reduce the risk of heart disease? 1.5+1=2.5

5. (a) Give evidences in favour of cyclic structure of D-glucose. 2.5
(b) Discuss a method to determine the ring size in D-glucose. 3
(c) Which two aldohexoses will form the same osazone? What are they called? 2
(d) What is invert sugar? How can you explain the phenomenon of inversion of cane sugar? 1+1.5=2.5
(e) Carry out the following conversions: 2+2= 4
(i) D-fructose to D-glucose
(ii) D-glucose to D-arabinose

6. (a) Discuss how the structure of alloxan has been established. 2
(b) Give a synthesis of uric acid from urea and ethylacetoacetate. 3
(c) What products would you expect to be formed when nucleic acids are hydrolyzed under different conditions? 3
(d) Describe briefly how the structure of nucleotide was established. 4
(e) Draw the structure of A-T and G-C pairs which have been observed in DNAs. 2

7. (a) Explain why some antibiotics are less active against Gram (-) bacteria than Gram (+) bacteria? 4
(b) Discuss the procedure of Gram staining differentiating Gram (-) bacteria than Gram (+) bacteria. 2
(c) The 'S' Sulphur in Penicillin is not in thiol form but it is in a five membered ring. Explain with necessary reactions. 3
(d) Outline the synthesis of Chloramphenicol. 2.5
(e) Write different mechanism of actions, an antibiotic can follow against bacteria. 2.5

8. (a) Define: DNA, RNA, Purine, Nucleoside 4
(b) Find out the structure of Uric acid by degradative study 3
(c) Explain that all amino acids exist as dipolar ions. 2
(d) Sketch the Whatson and Crick model of DNA showing different parts. 2
(e) Why nucleosides are non-reducing? 1
(f) How can you prove that Medicus structure of Uric acid is the actual structure? 2

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2020
Course Title: Chemistry of Natural Products
Course # Chem. 332, Full Marks: 70, Time: 4 hrs.

[Answer any five questions]

1. (a) Briefly discuss the isoprene rule and special isoprene rule with example. 2.5
(b) How would you chemically establish the following facts (give necessary chemical equations): 6.0
(i) Myrcene contains three carbon-carbon double bonds, two of which are in conjugation.
(ii) There is an α,β-unsaturated aldehyde system in citral molecule.
(iii) There is a methyl group with the β-carbon atom of the citral molecule with respect to the aldehyde group.
(iv) There is a primary alcoholic group in the farnesol molecule.
(v) Limonene has one double bond in position 8.
(vi) Farnesol is a primary alcohol.
(c) Give the biosynthesis of biogenetic isoprene unit from mevalonic acid. 2.0
(d) Write the structure elucidation of citral. 3.5

2. (a) What is mutarotation? Explain why maltose shows mutarotation but sucrose does not. 2.5
(b) Show the conformations of cellulose and starch and write the basic difference between them. 3.0
(c) Draw the structure of maltose sucrose cellulose, starch. 2.0
(d) How can you chemically distinguish between α-D-glucopuranose and β-D-glucofuranose. 3.0
(e) How will you prove that D(+)-glucose has reducing while methyl-D-glucoside has non reducing properties? 1.5
(f) How will you synthesize an aldohexose from an aldopentose? 2.0



3. (a) Write down the name and structure of nitrogeneous bases present in nuclic acids. 1.5
(b) Draw and point out the important features of the Watson and Crick model of the double helix structure of DNAs. 3.5
(c) What happens when allantoin reacts with (i) HNO_3, (ii) conc.HI at 100° C and (iii) alkali? 3.0
(d) Show the relation among Nucleic acids, Nucleosides and Nucleotides. 2.0
(e) Discuss the importance of purine bases. 2.0
(f) Write a short note on replication of DNA. 2.0

4. (a) Briefly discuss the extraction and purification methods of alkaloids. 2+2
(b) Write down the importance of Hofmann's exhaustive methylation method in Alkaloid chemistry with an appropriate example. 2.0
(c) What are the alkaloids of opium? What is the general test for detection of opium alkaloids. 2.0
(d) Write the synthetic methods of structure elucidation of penicillins. 3.5
(e) Write the clinical properties of chloramphenicol. 2.5

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5. (a) What do you mean by essential amino acids? Give the name and structure of two such amino acids. 1+1
(b) What is ninhydrin test? Write down the mechanism of the reaction of an α-amino acid with ninhydrin. 1+1.5
(c) Discuss the Edman's method for determining the sequence of α-amino acids in a polypeptide chain. 4.0
(d) How can you synthesize the tripeptide, Ala-Gly-Phe by using solid-phase peptide synthesis? Write down the advantages of this method. 3+1
(e) What is denaturation of proteins? 1

6. (a) How can you synthesize D-glucose from D-arabinose. 2.0
(b) Prove that D-glucose and D-fructose have identical configuration at C-3, C-4 and C-5. 2.0
(c) What is the significance of the letter D and the sign (+) in the name of D(+)-glucose? 1.5
(d) What is the basic differences between epimer and anomer. 2.5
(e) Write the methods for determining ring size in sugars. 3.0
(f) Describe the evidences for ring structure of aldoses. 3.0

7. (a) What do you mean by saponification number and iodine value of fat or oil? Write their importance in the analysis of fat or oil. 2+2
(b) What are phospholipids? Name the various types of phospholipids and draw their Chemical structures. 1+4
(c) Write down the biological functions of phospholipids. 3.0
(d) "HDL is better than LDL for human health" – explain. 2.0

8. (a) What are broad-spectrum antibiotics? Give the chemical names and structures of five natural penicillins. 1+2.5
(b) Complete the following reactions with structure. 3.0

$$\text{Penicillins} \xrightarrow{NaOH} ? \xrightarrow{-CO_2} ? \xrightarrow{H_2O-HgCl_2} ? + ?$$

(c) Give a synthesis of chloramphenicol. 3.5
(d) What do you mean by antibiotic resistance? Write down the strategies to prevent and control of antibiotic resistance by individuals. 1+3

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2018
Course Title: Chemistry of Natural products
Course No.: Chem. 332, Full Marks: 70, Time: 4 hours

Answer any five (05) questions

1. (a) What is mutarotation? Explain why maltose shows mutarotation but sucrose does not. 1+3
   (b) Prove that D-glucose and D-fructose have identical configuration at C-3, C-4 and C-5. 4
   (c) How can you synthesize D-arabinose from D-glucose? 2
   (d) Draw Haworth and conformational structure for α-D- glucopyrarose and β-D-glucopyranose. 4

2. (a) How do epimers and anomers differ? 2
   (b) What sugars are obtained by epimerizing C-2, C-3, C-4 of D-glucose respectively? 3+1
   (c) Explain how methylation and the degradation of the methylated products have helped in determining the size of the ring in D-glucose. 5
   (d) What is invert sugar? Mention the importance of invert sugar. 3

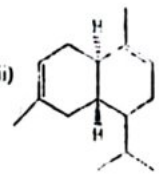
3. (a) What are the α-amino acids? Write down their general properties. 2
   (b) In general, in which configuration the natural α-amino acids exist? 0.50
   (c) Briefly discuss a general method for the synthesis of α-amino acids. 3
   (d) What is DCC? What role does it play in the formation of peptide? Explain with example. 2
   (e) Draw the partial *beta* (*β*)-pleated sheet structure of protein and answer the following questions: 2
      (i) Is it a primary or secondary structure of protein? 0.5
      (ii) When does the *beta* (*β*)-pleated sheet structure predominant? Give explanation in support of your answer. 1
   (f) What is isoelectric point? How would you calculate the isoelectric point of an α-amino acid having one amino and one carboxylic acid group? 1+2=3

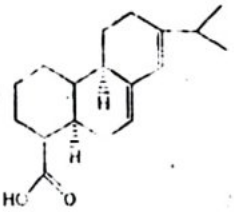
4. (a) What are alkaloids? How are they isolated from natural sources? 3
   (b) What products would you expect to be formed when ephedrine is treated with concentrated hydrochloric acid? 2
   (c) Discuss Hofmann's exhaustive methylation Method. 3
   (d) Morphine contains two hydroxyl groups and one of the two hydroxyl groups is phenolic in nature- Explain. 3
   (e) Outline the synthesis of tropic acid. 3

5. (a) What are purines? How is purine synthesized from uric acid? 2+2
   (b) Starting from ethylacetoacetate how would you synthesize uric acid? 3
   (c) Define the terms nucleosides, nucleotides and nucleic acid. 3
   (d) Discuss the double helix structure of DNA. 4

6. (a) How many isoprene units are present in the following compounds and also indicate the head to tail fashion? 4

(i) 

(ii) 

(iii) 

   (b) Explain the following facts. 3
      (i) Myrcene contains three carbon-carbon double bonds, two of which are in conjugation.
      (ii) Limonene has two double bonds and the second double bond is present between C-8 and C-10.
      (iii) Farnesol contains one primary alcoholic group.
   (c) How to synthesize citral from biological isoprene units? 4

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7. (a) What are antibiotics? Give one example.
   (b) Define broad spectrum and narrow spectrum antibiotics with suitable examples.
   (c) What product would you expect to be formed when penicillamine is treated with (i) (Br_2+H_2O) and (ii) Acetone.
   (d) How would you establish the following facts:
   (i) Chloramphenicol contains a *para*-disubstituted benzene nucleus.
   (ii) The carboxyl group in penicillamine is the carboxyl group in penicillin itself.
   (iii) There is an isopropyl end-group in penicillamine.
   (e) Show that there are two hydroxyl groups in the chloramphenicol molecule and also show that the two hydxoyl groups are alcoholic in nature. 2

8. (a) What are lipids? Give example. 1
   (b) Briefly discuss the general classification of lipids? Give example for each class with structures. 2
   (c) What are phospholipids? Give example. 1
   (d) How would you classify phospholipids on the basis of the structural backbone? Give examples for each class with structures. 2
   (e) What are steroids? What is the basic structural skeleton of steroids? Draw the chair form of the structural skeleton of steroids. 1+1+1=3
   (f) Explain the blanc's rule with suitable examples. 2
   (g) Using blanc's rule how can you determine the ring size A and D of cholesterol. 3

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Department of Chemistry
Jahangirnagar University
B.Sc. (Honours) Part III Examination, 2017
Course Title: Chemistry of Natural Products
Course No. Chem. 332

Total Marks: 35 Time : 2.5 Hours

[Answer any FIVE questions]

1. (a) Define the terms glycosidic linkage, anomer and epimer. 3
(b) How will you prove that D(+)-glucose has reducing while methyl-D-glucoside has non-reducing properties? 2
(c) How will you synthesize an aldohexose from an aldopentose? 1.75
(d) What is the significance of the letter D and the sign (+) in the name of D(+)-glucose? 2

2. (a) Write a note on "Mutarotation". 2
(b) Glucose and fructose give identical Osazones – explain. 1.75
(c) Show the conformations of cellulose and starch. 2
(d) Describe the methylation method to determine ring size of glucose. 3

3. (a) What are essential amino acids? Why are they so called? 1.75
(b) Give two methods for the preparation of α-amino acids. 2
(c) Write the structures of all possible dipeptide obtainable from glycine and alanine. 2
(d) What are proteins? Discuss the α-helix structure of proteins. 3

4. (a) Draw the conformational structure of sucrose and maltose and answer the following questions: 2
(i) What type of linkage present between the D-glucose and D-fructose units in the sucrose molecule? 0.5
(ii) Why does sucrose not undergo mutarotation? 1
(iii) What is invert sugar? Why is invert sugar sweeter than sucrose? 2
(iv) Indicates the anomeric carbon atom of maltose. 0.5
(v) Is maltose a reducing sugar? Give explanation in support of your answer. 1.75
(b) Draw the chair form of α-D-glucopyranose and α-L-glucopyranose 1

5. (a) Briefly discuss the isoprene rule and special isoprene rule with example. 1.75
(b) How would you chemically establish the following facts (give necessary chemical equations): 4
(i) Myrcene contains three carbon-carbon double bonds, two of which are in conjugation.
(ii) There is an α,β-unsaturated aldehyde system in the citral molecule
(iii) There is a methyl group with the β-carbon atom of the citral molecule with respect to the aldehyde group.
(iv) There is a primary alcoholic group in the farnesol molecule.
(c) Give a synthesis of farnesol. 3

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6. (a) Draw the structure of morphine and answer the following questions (give explanation in support of your answer).
(i) Show that the nitrogen atom of morphine is present in the ring system, and also show that there is a methyl group with the nitrogen atom of morphine.
(ii) There are two hydroxyl groups in the morphine molecule of which one is alcoholic in nature.
(iii) There is a phenanthrene nucleus in the morphine molecule.
(b) What product would you expect to be formed when atropine is warmed with aqueous $Ba(OH)_2$ solution? Draw the structures of the expected products.
(c) Outline all the steps involved in the synthesis of ecgonine from tropinone.

7. (a) What are purine bases? Discuss their importance.
(b) Uric acid on oxidation with alkaline permanganate yields allantoin and CO_2. Discuss how the structure of allantoin is derived.
(c) Show the relation among Nucleic acids, Nucleosides and Nucleotides.
(d) Discuss the replication of DNA.

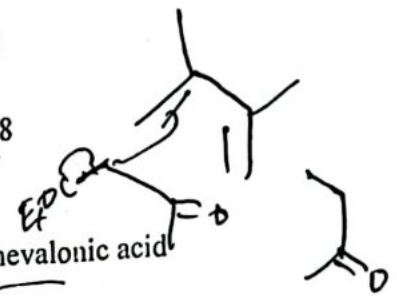
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Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2016
Course Title: Chemistry of Natural Products
Course No: Chem - 332
Time: 2.5 Hours, Total Marks: 35

Answer any four questions

1. (a) What are terpenoids? How are they classified? 1+1.5
   (b) Establish the following facts: 1.5+1.5
   (i) Limonene has one double bond in position 8
   (ii) Farnesol is a primary alcohol.
   (c) Give a synthesis of limonene from isoprene 2
   (d) Show the biosynthesis of isoprene unit from mevalonic acid 1.25



2. (a) What is mutarotation? Explain the phenomenon in the case of D-glucose. 2
   (b) Discuss a method to determine the ring size in D- glucose. 2.5
   (c) What is invert sugar? How can you explain the phenomenon of inversion of cane-sugar? 2
   (d) How can you synthesize D-glucose from D-arabinose? 2.25

3. (a) Which amino acid is not chiral? Draw its structure. 1
   (b) Show the reaction of ninhydrin with phenylalanine. 1.5
   (c) A tripeptide is composed of L-valine, L-tyrosine and L-alanine (Val.Tyr.Ala)- Write its structure. 1.5
   (d) What reagent is used in the Erdman degradation for the N-terminal group analysis of Peptides? Show the reaction. 1.75
   (e) Which type of bonding is predominantly involved in the formation of beta-pleated sheet structure of proteins? 1
   (f) Explain the relationship between genes and polypeptides. 2

4. (a) Give a synthesis of uric acid from urea and ethylacetoacetate. 2
   (b) Write down the name and structure of nitrogeneous bases present in nucleic acids. 2.5
   (c) Draw and point out the important features of Watson and Crick model of the double Helix structure of DNAs. 3.25
   (d) Give an outline on the replication of DNAs. 1

5. (a) What are the alkaloids of opium? What is the general test for detection of opium alkaloids? 1+1
   (b) Draw the structures of morphine and explain how you would establish the fact that morphine contains a phenanthrene nucleus. 0.75+1
   (c) Heroin is chemically derived from morphine. Describe the chemical reaction that converts morphine to heroin. 1

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(d) How can you detect the following groups?
(i) >N-methyl group in ephedrine?
(ii) Phenolic –OH in morphine?
(e) What is the source of ephedrine? Briefly describe one suitable method for extraction of ephedrine.

6. (a) What is starch? State which structural feature is the key to the colour change in the iodine test for starch? 2
(b) Why humans are unable to digest cellulose? 1+1
(c) What are pentoses? What are the roles of pentoses in DNA and RNA molecules? 1
(d) What are the three main types of ribonucleic acids? Mention the function of each of them. 1+1.5
(e) How many nucleotides are required to code for a specific amino acid? Give one such example. 0.75+1.5

7. (a) Establish any four of the following statements: 1
(i) On ozonolysis 1 mole of myrcene gives 2 moles of formaldehyde, 1 mole of acetone and one mole of a kitodialdehyde, $OHCCH_2CH_2COCHO$. 2x4
(ii) D-glucose and D-fructose have identical configuration at C-3, C-4 and C-5.
(iii) Tropine contains a pyrrolidine ring in its structure.
(iv) All amino acids exist as dipolar ions.
(v) Ribose or deoxyribose is in furanose form in the structure of nucleic acids.
(vi) Citral showed positive tests with 2,4-DNPH as well as Tollen's reagent.
(b) Draw the conformational structure of α-anomer of (+)-maltose. 0.75

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Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2015
Course Title: Chemistry of Natural Products
Course No: Chem - 332
Time: 2.5 Hours, Total Marks: 35

Answer any four questions

1. What are isoprene and special isoprene rule? 2.0
   What product would you expect to be formed when limonene is treated with (i) dil. H_2SO_4 and (ii) NOCl? 2.0
   Outline a synthesis of myrcene. 2.0
   How can you prove that camphor contains a $-CH_2CO-$ group? 2.0
   What do you mean by essential oils? 0.75

2. Show the conformational structure of α-D-glucopyranose and β-D-glucopyranose. 1.0
   How would you distinguish chemically between α-D-glucopyranose and β-D-glucopyranose? Give necessary equations. 3.0
   Write the mechanism for the base catalyzed conversion of D-fructose into D-glucose and D-mannose. 3.0
   What are the basic differences between anomers and epimers? 1.75

3. (a) What are the main characteristics of plant alkaloids? 1.0
   (b) How can you detect the following groups in an alkaloid?
   (i) phenolic –OH (ii) $-N(CH_3)_2$ 2.0
   (c) Discuss the extraction and isolation of alkaloids from plants. 2.5
   (d) Give a synthesis of tropine from succinaldehyde, $OHCCH_2CH_2CHO$. 1.5
   (e) How can you establish the fact that morphine contains a phenanthrene nucleus? 1.0
   (f) Indicate the medicinal uses of D(-)-ephedrine. 0.75

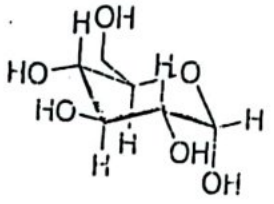
4. What do you mean by essential amino acids? Give the name and structure of two such amino acids. 2.0
   What is ninhydrin test? Write down the mechanism of the reaction of an α-amino acid with ninhydrin. 2.5
   Briefly discuss the Edman's method for determining the sequence of α-amino acids in a polypeptide chain. 2.5
   What is denaturation of protein? What effect does it have on properties of a protein? 1.75

5. Calculate the double bond equivalent value of myrcene. What theoretical conclusion would you draw about the structure of the myrcene molecule from the calculated double bond equivalent value? 2.0
   How would you chemically establish the following facts (give necessary chemical equations). 4.0
   Myrcene contains three $C=C$ bonds, two of which are in conjugation.
   Myrcene is an open chain compound.
   There is an α,β-unsaturated aldehyde system in the citral molecule.

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6. (a) Discuss how the structure of alloxan has been established? 2.
(b) Give a synthesis of guanine from uric acid. 2.
(c) What products would you expect to be formed when nucleic acids are hydrolyzed under different conditions? 2.
(d) How can you differentiate ribonucleotide-5'-phosphate from its isomers 2'-and 3'-phosphates? 1.0
(e) Write down the differences between RNAs and DNAs. 1.7

7. (a) What are the basic differences between proteins and nucleic acids? 1.2
(b) Draw the structures of adenosine 5'-monophaospahate and 2'-deoxyadenonisine 5'-monophasophate. How can you distinguish between them chemically? 3.0
(c) Write down the major product of the following reactions and give reason in favour of your answer. 3.0



α-D-Glucopyranose

$(CH_3CO)_2O$ (in excess), Pyridine, 0°C → ?

$(CH_3CO)_2O$ (in excess), Pyridine, 100°C → ?

(d) Explain why acetals undergo hydrolysis more rapidly than ordinary ethers. 1.5

